

Gina Sohn

✉ ginasohn@stanford.edu | 🌐 Website | 🔗 LinkedIn

Updated September 24, 2026

RESEARCH INTERESTS

HW/SW Co-design, Compiler, Domain-specific Language, Hardware Accelerator

EDUCATION

Stanford University

Sep. 2022 – Present

Ph.D., Electrical Engineering

Advisor: **Kunle Olukotun**

Seoul National University

Mar. 2018 – Aug. 2022

Bachelor of Science, Computer Science and Engineering

Total GPA of 4.16 / 4.3 (Summa Cum Laude)

PUBLICATIONS

Sieve: Dynamic Expert-Aware PIM Acceleration for Evolving Mixture-of-Experts Models

Jungwoo Kim, Rubens Lacouture, Genghan Zhang, [Gina Sohn](#), Qizheng Zhang, Swapnil Gandhi, Christos Kozyrakis, Kunle Olukotun

arXiv preprint, May 2026 (Under review) [\[paper\]](#)

Streaming Tensor Program: A streaming abstraction for dynamic parallelism

[Gina Sohn](#), Genghan Zhang, Konstantin Hossfeld, Jungwoo Kim, Nathan Sobotka, Nathan Zhang, Olivia Hsu, Kunle Olukotun

Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Mar. 2026 [\[paper\]](#)

The Dataflow Abstract Machine Simulator Framework (🏆 ISCA Distinguished Artifact Award)

Nathan Zhang, Rubens Lacouture, [Gina Sohn](#), Paul Mure, Qizheng Zhang, Fredrik Kjolstad, Kunle Olukotun

Proceedings of the International Symposium on Computer Architecture (ISCA), June 2024 [\[paper\]](#)

ARK: Fully Homomorphic Encryption Accelerator with Runtime Data Generation and Inter-Operation Key Reuse

Jongmin Kim, Gwangho Lee, Sangpyo Kim, [Gina Sohn](#), John Kim, Minsoo Rhu, and Jung Ho Ahn

Proceedings of the 55th IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct. 2022 [\[paper\]](#)

Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs

Shine Kim, Yunho Jin, [Gina Sohn](#), Jonghyun Bae, Tae Jun Ham, and Jae W. Lee

Proceedings of the 19th USENIX Conference on File and Storage Technologies (FAST), Feb. 2021 [\[paper\]](#) [\[slides\]](#)

WORKSHOP

Dynamic Memory for Distributed Architectures

[Gina Sohn](#), Genghan Zhang, Suguna Velury, Nathan Zhang, Kunle Olukotun

The Sixth Young Architect Workshop (Organized in conjunction with ASPLOS 2024) [\[pdf\]](#)

HONORS AND SCHOLARSHIPS

Graduate Research Fellowship Program, National Science Foundation

2025–2028

Stanford Graduate Fellowship (Burt & Deedee McMurtry Fellow), Stanford University

2022–2025

National Scholarship for Science and Engineering, Korea Student Aid Foundation

2020–2021

Eminence Scholarship, Seoul National University

2019

ACADEMIC SERVICE

Student Program Committee Member, IEEE/ACM International Symposium on Microarchitecture

2026

RESEARCH EXPERIENCE

Pervasive Parallelism Laboratory

Stanford University

Advisor: Kunle Olukotun

Apr. 2023 – Present

Sep. 2022 – Dec. 2022

Stanford Accelerate Laboratory

Stanford University

Advisor: Priyanka Raina

Jan. 2023 – Mar. 2023

SCALable Computer Architecture Laboratory

Graduate School of Convergence Science and Technology, Seoul National University

Advisor: Jung Ho Ahn

Jan. 2022 – Apr. 2022

Architecture and Code Optimization Laboratory

Samsung-SNU Cooperation R&D Center, Seoul National University

Advisor: Jae W. Lee

Jul. 2020 – Sep. 2020

TEACHING EXPERIENCE

Digital Systems Design Lab, Stanford University

Teaching Assistant

Apr. 2024 – Jun. 2021

- Topics covered: Software / hardware design tradeoffs, Algorithm design for pipelining and parallelism, System latency and throughput tradeoffs, FPGA optimization techniques.

Computer Programming, Seoul National University

Teaching Assistant

Sep. 2021 – Dec. 2021

- Covered C++ and Java programming, Object-Oriented Programming
- Revised lab materials and assignment descriptions, Participated in weekly Professor-TA meetings, Administered laboratories for 170 students every week, Answered questions online

EXTRACURRICULAR & LEADERSHIP ACTIVITY

Stanford Women in Electrical Engineering (WEE)

Sep. 2023 – Present

Seoul National University College of Engineering Honor Society (STEM)

Academic Activity Director (Sep. 2021 – Feb. 2022)

Mar. 2021 – Jun. 2022

Korean-American Women in Science and Engineering (KWise)

2021 Mentee Team Leader (Chapter: Los Angeles, USA)

Mar. 2021 – Dec. 2022

Seoul National University Security Research Group (Guardian)

Mar. 2019 – Jul. 2019